communications earth & environment

Article

A Nature Portfolio journal

Article in Press

https://doi.org/10.1038/s43247-026-04080-z

Fluid inclusions establish natron as a key astrobiological target on Enceladus

Check for updates

Liam J. Perera [1] ✉, James Le Houx [2,3,4], Alberto Leonardi[1,5,6], Sarah J. Day[1] & Stephen P. Thompson [1]

Saturn's moon Enceladus is a priority target for both NASA Flagship and ESA L4 class missions, as it possesses the three essential ingredients for life: liquid water, a source of energy, and all essential (CHNOPS) elements. As Enceladus's ice crust is geologically active, future missions must understand its habitability through the lens of cryogeochemistry; yet the phase behaviour of Enceladus-type solutions is poorly constrained. Here, we investigate the freezing behaviour of an Enceladus-type Na-Cl-CO_3 solution using combined synchrotron X-ray computed microtomography and diffraction. We report the observation of fluid inclusions in cryogenic natron ($Na_2CO_3.10H_2O$), a mineral predicted to occur on and within Enceladus. We show that natron precipitation traps and preserves pockets of parent fluid, providing a window into the geochemistry of Enceladus's interior and establishing natron as a mineral of high astrobiological importance. Consequently, we argue that future missions must have the capability to identify natron at the surface and that direct sampling should prioritise fresh plume deposits where natron is abundant due to its potential to trap and preserve signatures of life.

Enceladus, a moon of Saturn, displays geophysical and geochemical evidence of recent and ongoing geological activity[1–3]. The most dramatic evidence of activity is the presence of distinct jets, feeding a large plume that emanates from the South Polar Terrain (SPT)[2,4,5]. These plumes contain ice[5], volatile gases[6], inorganic salts[7] and evidence of all six CHNOPS elements[8,9]. Multiple hypothetical plume mechanisms connect the source of these plumes with a subsurface liquid water ocean[7,10,11]. Consequently, Enceladus is a prime location for the search for life beyond Earth and is the designated target for both a NASA Flagship-class and an ESA-L4 mission.

Direct sampling of Enceladus's ocean is unlikely, as the closest estimates place it at least several kilometres beneath the surface[10,12]. However, transport processes in the crust can move ocean-derived material across the solid-liquid phase boundary, bringing it to the surface[13]. Samples can therefore be taken from the surface or the plumes to reveal geochemical information about the ocean below. The primary science objective of future Enceladus missions is to look for signatures of life on the surface[14,15]. The low pressure and radiation environment on the surface are likely damaging but not limiting to such detection[15,16]. An open question is which inorganic signatures could be used to prioritise site selection for surface characterisation. The potential for certain salts to preserve or shield biological material would be a useful discriminator.

When an aqueous solution cools and solidifies, the solubility limits of the dissolved components are exceeded and they precipitate out as solids. As Enceladus's surface temperature is on average 70 K[17], surface detection of ocean signatures will be of solids or materials hosted in solid matrices. The solid phases in a frozen sample, provided they are not exogenously deposited, would originate from the dissolved components in the parent fluid. Communication of ocean geochemistry to the surface would range from rapid plume transportation to slow crustal overturning. The solid-phase geochemistry, texture and petrology, within a surface sample could be used to reconstruct the ocean chemistry along with the physiochemical evolution during transport and therefore habitability.

One such microstructural feature of importance is fluid inclusions, trapped pockets of parent fluid preserved in a solid matrix. The main relevant low temperature environments where fluid inclusions are observed are in evaporitic salt deposits[18–21] and in carbonate speleothems[22]. Furthermore, salt precipitation is of astrobiological significance as several evaporitic salt deposits have been found with viable microbes trapped within fluid inclusions[23–26].

Cryogenic precipitation differs from evaporitic precipitation due to the temperature regimes, physical processes and resulting precipitation sequences. Cryogenic precipitation produces hydrated minerals that often grow within the confinement of ice. The evaporation of sea water produces the cubic, anhydrous mineral halite (NaCl), which can form hopper-like crystals that grow unconfined allowing them to trap fluid inclusions[18,27,28]. The freezing of sea water produces hydrohalite ($NaCl.2H_2O$) at −21 °C, a

[1]Diamond Light Source, Rutherford Appleton Laboratory, Didcot, UK. [2]Old Royal Naval College, University of Greenwich, London, UK. [3]ISIS Neutron & Muon Source, Rutherford Appleton Laboratory, Didcot, UK. [4]The Faraday Institution, Harwell Science and Innovation Campus, Didcot, UK. [5]Technische Fakultät, Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen, Germany. [6]Department of Earth and Atmospheric Sciences, Indiana University, Bloomington, IN, USA. ✉e-mail: liam.perera@diamond.ac.uk

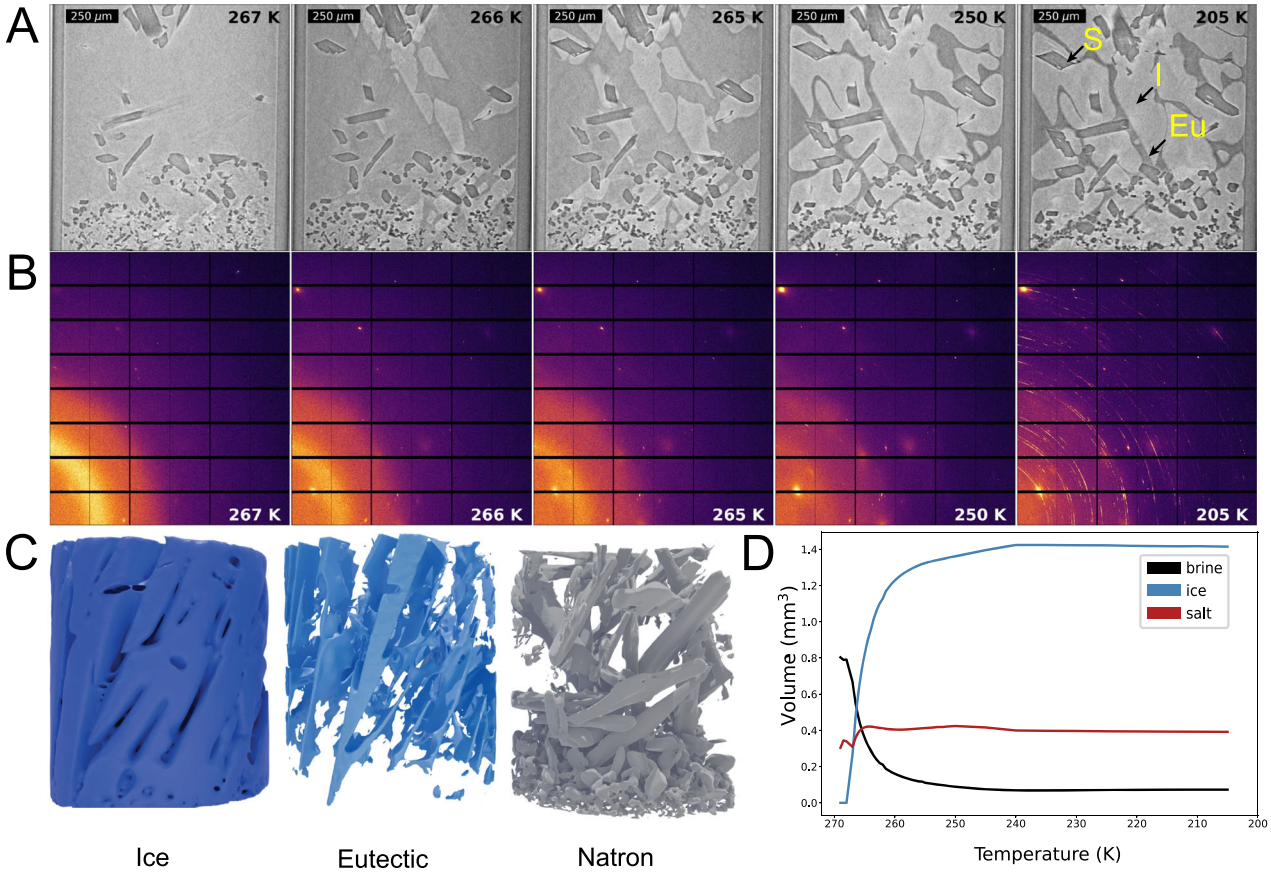


Fig. 1 | Identification of major phases and structure during real time freezing of an Na-Cl-CO_3 solution. A Reconstructed XY slice vs temperature showing the growth of each phase and development of the final microstructure. Series begins at the onset of ice formation at 267 K .Natron (s), ice (i) and eutectic (Eu). **B** Corresponding 2D diffraction patterns showing texture evolution **C** Rendered volumes of ice, the eutectic phase, and natron in final scan at 205 K **D** Plot of evolution of volume of each phase with temperature calculated from segmented volumes.

monoclinic mineral that forms flat, pseudo-hexagonal crystals that are confined within ice inclusions and brine channels[29]. The direct study of cryogenic salts is difficult due to the narrow stability envelope within terrestrial conditions, that limits their preservation within the geological record. Therefore, their precipitation behaviour must be explored experimentally.

Enceladus's chemistry is carbonate dominated[7,30], as is that of several other icy bodies: Bennu[31], Ryugu[32], Ceres[33] and Ariel[34]. Because carbonates have low solubility at low temperatures[35], they are some of the first cryogenic minerals to form during freezing. In a cryogenic precipitation sequence, early phases would trap less-evolved ocean water, whereas a later mineral would preserve concentrated, chemically evolved brine. This makes carbonates a high-value record of primary ocean fluids if they can form inclusions in cryogenic settings. However, terrestrial evidence of carbonate hosted inclusions is limited to inorganic[22] and biogenic calcite in unconfined settings[36].

If inclusions form in cryogenic carbonates and they are present on Enceladus's surface, these inclusions and any potential biological material would be protected from the deleterious effects of the space environment. Given the importance of carbonate chemistry on Enceladus, we explore the freezing behaviour of an Enceladus-type, Na-Cl-CO_3 solution in real time using combined synchrotron X-ray computed micro-tomography (μCT) and diffraction (XRD). We show the widespread presence of primary fluid inclusions in cryogenic $Na_2CO_3.10H_2O$ (natron). We discuss the implications for fluid transport and preservation on Enceladus and other carbonate bearing icy bodies. We propose that direct sampling missions should target these salts due to their potential to trap ocean fluids or biosignatures if they are present.

Results and discussion

Cryogenic phase evolution and texture

Time resolved tomography and diffraction were collected on K11-DIAD (Diamond Light Source, UK) during the freezing of an Na-Cl-CO_3 solution (1 mol kg^{-1} NaCl, 0.5 mol kg^{-1} Na_2CO_3). K11 operates with two independently configurable X-ray beams that allow the collection of spatially correlated imaging and diffraction data. This study revealed the solidification of an Na-Cl-CO_3 solution using time-resolved tomography (Fig. 1A) and diffraction (Fig. 1B), and identified the solid phases present using the diffraction data.

Three solid phases were identified and segmented from the imaging data of the final frozen sample: hexagonal ice (Fig. 1C Left); the eutectic phase, comprised of ice, natron and hydrohalite (Fig. 1C Middle); and individual natron crystals (Fig. 1C Right). Due to the presence of motion artefacts from 270 to 269 K, segmentation was only reliably achieved in temperature points below 268 K.

Hexagonal ice was the dominant solid phase, making up 75% of the imaged volume. The basal plane represents a slow-growth face of ice, and dendritic growth proceeds preferentially along the <100> and <112> directions[37,38]. This was evident in our diffraction data as an under-representation of basal reflections (Supplementary Fig. 2) and the spotty, single-crystal reflections in our raw diffraction data (Fig. 1B). Our imaging data revealed that the ice grew dendritically down the axis of the capillary with the direction of cooling (Fig. 1A, C). As a result, the dimensions and aspect of the crystals in the plane of the beam coupled with the small beam size produce strong single crystal reflections. We therefore limited our diffraction analysis to phase identification with Pawley refinement[39], which

is intensity independent. The final quenched sample was assumed to be representative of all phases present.

The eutectic phase, a mixture of ice and hydrohalite and natron, made up 4% of the imaged volume. This represents the solidified brine network, of eutectic composition, present in the ice mass. Of the eutectic volume, 3.8% was characterised as disconnected. Individual structure within the eutectic was not resolvable with imaging. The appearance of rings from a previously spotty pattern at 245 K in the raw diffraction was caused by the formation of multiple small crystallites (Fig. 1B). The scattering signal from hydrohalite appeared at this eutectic temperature and its powder-like scattering (clear rings) suggests it precipitated as a fine-grained mass.

The final major solid phase was subhedral crystals of sodium carbonate in the form of natron. This phase made up 21% of the imaged volume. Similarly to the ice crystals, the small beam and crystal dimensions produced spotty single crystal reflections. DIAD's geometry only interacts with a reduced spherical arc of the Ewald sphere and the sample does not spin during collection, preventing the reconstruction of a complete intensity profile. The presence of natron is more evident in individual diffraction scans at 205 K (Supplementary Fig. 3). This was due to a population of randomly oriented, smaller crystals at the bottom of the imaged volume producing a more powder-like signal.

In our imaged volume there were two separate habits of natron that formed over time. Large subhedral grains precipitated and grew before ice formation. After the onset of ice formation, the subhedral carbonate crystals remained trapped within the mass of ice and their elongated profile prevented further movement. The presence of large subhedral crystals suggests that they grew unconfined, and once contained by ice, the growth is inhibited, which can be seen in the salt phase volume in Fig. 1D. The initial growth of the ice leads to a temporary interruption to the growth of the natron. Once the ice encapsulated large grains, smaller secondary offshoots of natron formed from larger crystals and were templated by the ice structure (Supplementary Fig. 4). These secondary precipitation events contain defects and holes that have the potential to form inclusions (Supplementary Fig. 5).

Fluid entrapment in cryogenic natron

In this Na-Cl-CO_3 system, we found the widespread presence of fluid inclusions within the subhedral natron crystals (Fig. 2A). This is, to the best of our knowledge, the first time that fluid inclusions have been observed non-destructively in cryogenic salts. μCT imaging combined with XRD provides a non-destructive way to identify the microstructure of the natron in-situ.

The inclusions identified were a mix of primary, intragranular and pseudo-inclusions (cavities) (Fig. 2C). As the inclusion sizes were close to the limits of our imaging resolution, we were unable to observe their freezing. We would expect that these inclusions would preserve the parent solution at the timing of enclosure and once frozen would be a mixture of ice, hydrohalite and natron. As all inclusions formed during the growth of the host grain, they can be classified as primary inclusions (Fig. 2B). No recrystallisation of natron was observed.

Larger salt grains were more likely to contain inclusions and would contain larger inclusions (Fig. 2D). Of the separated salt grains ($n = 3779$), 3% contained fluid inclusions. However, within grains larger than 1×10^{-4} mm^3 58% contained inclusions and 100% contained inclusions in grains above 1×10^{-3} mm^3 ($n = 29$). It is important to note that the particle size distribution (Supplementary Fig. 6) and consequently our size sampling dominates this observed relationship as there were fewer large crystals in our imaged volume.

Natron crystallisation on Enceladus

Our experiment demonstrates fluid inclusion formation in a cryogenic salt. Our objective was not to reproduce the exact composition of Enceladus's ocean water, but to establish whether cryogenic natron is capable of forming and trapping fluid inclusions under freezing conditions relevant to icy worlds. The crystallisation sequence observed here is governed primarily by the thermodynamic phase behaviour of the Na-Cl-CO_3-H_2O system and is therefore expected to be broadly transferable to Enceladus. In contrast, crystal size, morphology, preferred orientation and the abundance of inclusions are controlled by kinetic factors including cooling rate, concentration, confinement and thermal gradients. These textural properties should therefore be interpreted qualitatively rather than quantitatively with respect to Enceladus. We reason from our observation toward the likely formation within two regimes: the plume and the crust.

Natron formation is consistent with the expected phase behaviour of the system at these temperatures[40]: owing to its low solubility, natron precipitates early in the cryogenic sequence, before the chloride phases, giving it a high potential to be enclosed within growing ice. As Na, Cl and CO_3 species have all been detected in the plume[7,30], we expect that natron would form on and within Enceladus.

In our experiment, large natron grains nucleate and grow before bulk ice formation (Fig. 1C), so the fluid they enclose is close to the original solution composition. At bulk-ocean salinity, by contrast, ice is expected to form first and natron to precipitate later, from a residual liquid already concentrated by solute exclusion at the advancing ice front; inclusions trapped in this regime would preserve an evolved, fractionated brine rather than near-primary ocean water. Fractionated-brine inclusions concentrate dissolved species and record the physiochemical pathway of freezing. In settings in which natron precipitates early, they would preserve the least-fractionated fluid and are therefore the highest-value targets for reconstructing ocean chemistry.

In a bulk-ocean scenario, natron would precipitate within brine channels in the ice, forming structures similar to the secondary precipitation events we observed. These confined, ice-templated growths contained inclusion-like defects (Supplementary Fig. 5), suggesting that channel-confined crystallisation can also trap fluid, although inclusion formation is likely to be less frequent than in unconfined growth and our data show that inclusion occurrence increases strongly with grain size (Fig. 2D).

The petrology of natron can be used to reconstruct its evolution. Large crystals would be evidence of slow, unconfined growth, while the skeletal, fine-structured, secondary precipitation structures we see are evidence of confined growth as they are templated by the ice. In a more complex aqueous system, as we may expect of Enceladus's ocean, the microscale context of a mineral's crystallisation can be used to construct a more comprehensive view of the evolution of a cryogenic system.

Plume transport and surface deposition

Enceladus provides access to interior geochemical information through the eruption of sustained cryovolcanic plumes around its southern pole[1,2]. The exact mechanism by which the material reaches the surface is not known but the species within the plume require that some portion of the material originates from dispersion of a liquid source[9,30,41].

Within the plume, the crystallisation of natron will be controlled by the starting fluid composition, droplet size and vent temperature and pressures. The model that best explains observations is that tidal flexure forces the opening of cracks, exposing ocean water and volatiles to space[42–44]. Boiling and exsolution of the liquid source create dispersed droplets that are cooled and freeze as they are transported by an accelerating gas phase[45]. The plume erupts as a mixture of sub-mm solid particles and gas[45]. During this process, droplets pass from liquid to solid. Although the precise exit temperature is not known, surface temperatures near the plume vents are observed to be anomalously warm (140–225 K) compared to the average (70 K)[17,46,47]. Although we did not explore conditions down to surface temperatures, this work shows that much of the solidification is complete by 250 K.

Evaporative cooling is the expected dominant cooling mechanism within the plume and cooling rates for independent droplets will be dependent on their size and the vent conditions, with smaller droplets cooling faster[48]. Although vent conditions are unknown, our experimental cooling rate sits at the lower end of estimated conditions that could range from 5 to 10^6 K min^{-1} [49]. Dendritic and skeletal growth is more likely in faster precipitating solids, leading to more inclusions[18], however, in small

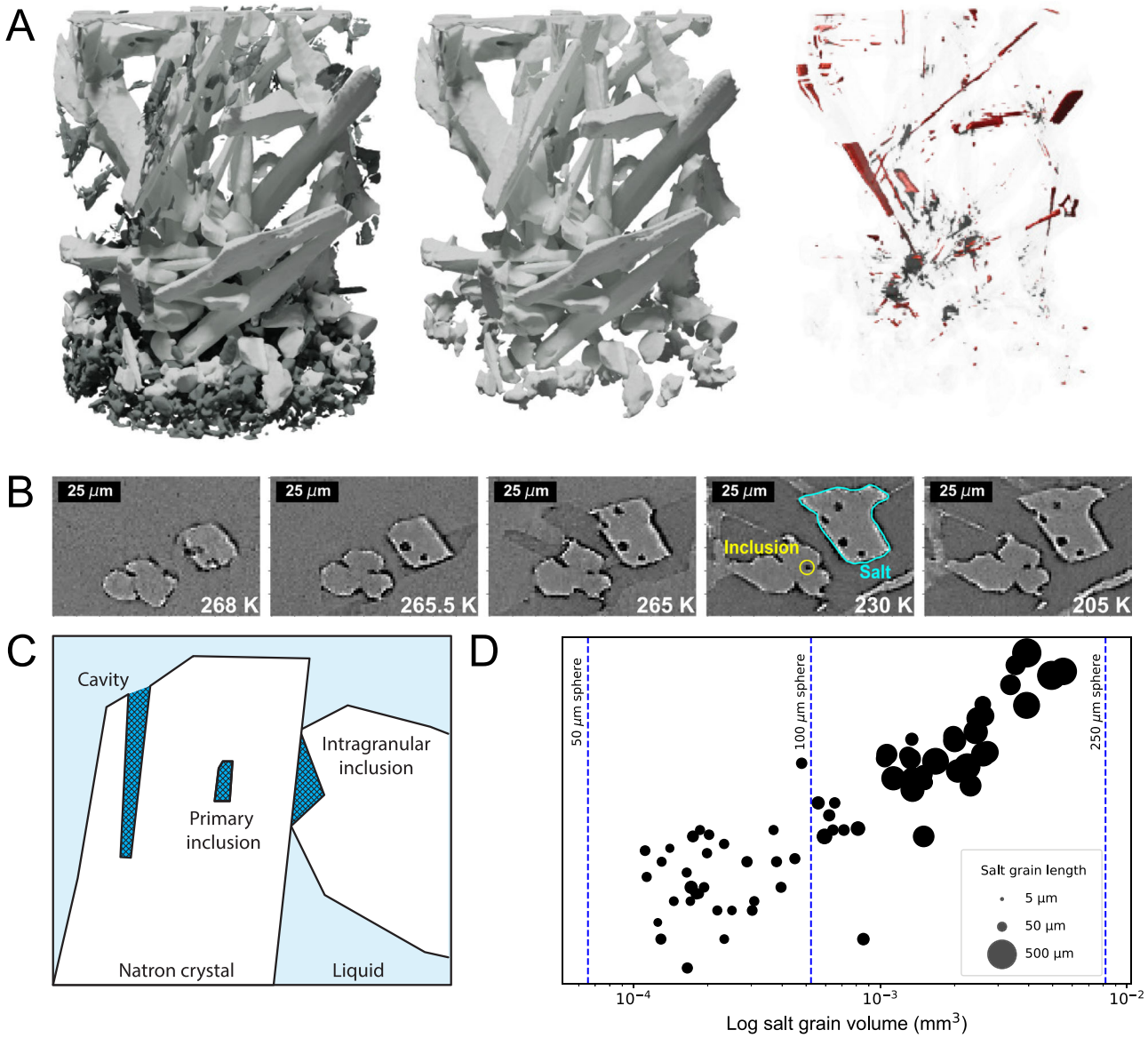


Fig. 2 | Analysis of precipitated natron and its inclusions. A Rendering of segmented salt particles. Particles with a volume above 1×10^{-3} mm^3 are displayed in white, with their associated inclusions shown in red. Salts not included in the analysis are shown as brown. **B** Temperature series showing formation of primary fluid inclusions within natron from reconstructed XY slices. A natron crystal (blue) and an associated inclusion (yellow) are labelled **C** Schematic representation of the types of inclusions we observed in the natron crystals. **D** The size of the host grain versus the size of its associated inclusion. The size of each point represents the long axis length of the natron crystal. The volume of equivalent spheres are displayed as dashed lines for reference.

droplets at fast cooling rates particles may vitrify when solidifying. In very cold and fast cooling regimes, droplets can vitrify and can partially vitrify at cooling rates as low as 100 K min^{-1} [50]. Fast cooling and vitrification of droplets would inhibit natron crystallisation and could lead to alternative, disordered and metastable phases[51,52]. Our experiment sits within the lower range of plausible plume cooling rates and suggests that inclusions within natron should form within larger plume droplets due to the slower cooling rate and reduced confinement, and that the early vent conditions will strongly control the crystallisation behaviour of natron.

The effect of confinement prevents the further growth of natron, suggesting that the diameter of a dispersed droplet would place an upper limit on the size of the precipitating natron crystals. The largest long-axis of the natron crystals we observe is 485 μm and the smallest is 34.3 μm, although our analysis of crystals was limited to particles larger than 0.0001 mm^3 (Supplementary Fig. 6). The cooling rate of a dispersed droplet would further control the size of the natron crystal, with larger droplets cooling more slowly, leading to larger natron crystals. Our final imaged volume was 0.91 mm^3, which is equivalent to a sphere of approximately 1.2 mm in diameter while the longest crystal axis we observed was 485 μm. As the long axis of this largest crystal was oriented at 38 degrees, it is more appropriate to set the long axis of our volume (1.2 mm) as the confining length (Supplementary Fig. 7). Therefore, we may expect that large crystals would require droplet diameters closer to that of the long axis of our imaged volume. We designed the experiment so that the cooling direction was down the axis of the capillary to induce directional growth meaning radial confinement is only affecting the slow growth axis of the ice phase.

Cassini measurements of particles within the E-ring suggest that the upper salinity estimate of the source is 0.2 mol kg^{-1}Cl and 0.1 mol kg^{-1} CO_3[30]. However, a mushy, more concentrated source for the plumes is physically possible[53], and the plume material may be diluted by vapour condensation[45,53]. Additionally, Cassini measurements are based on the upper plume (closest fly by at 25 km) and E-ring particles which were dominated by particles of micrometre to nanometre in size[54,55]. Larger salt-rich grains may be discriminated against in the upper plume. Remote

observation suggests larger surface grain sizes can be found closer to the plume sources[56], and gas-particle acceleration models suggest that particles of up to 2 mm could be ejected close to the plume vents[57]. We therefore hypothesise that large, salt-rich ice grains, those most likely to contain natron and its inclusions, are preferentially deposited close to the plume sources.

Crustal transport and cryo-concentration

The size of the crystals we observe is likewise not unique to high-salinity solutions. Large salt crystals of ikaite ($CaCO_3.6H_2O$) up to 700 μm [58] and hydrohalite up to 100 μm [29] have been observed in sea ice at millimolal ocean salinity, where slow cooling and the persistence of liquid above the eutectic allow ripening to dominate growth. Large crystals can form even from dilute fluids, provided cooling is slow; this regime is most relevant to the subsurface and tectonic pathway below rather than to rapid plume-droplet freezing. Slower cooling allows crystals grow in thermal equilibrium and surfaces can recrystallise, reducing surface defects and therefore inclusion formation. In a bulk ocean scenario where ice-growth rates are likely to be sub-mm $year^{-1}$ [59], inclusion formation will be reduced.

In slowly cooled, cryo-concentrated subsurface settings, larger natron crystals could form and subsequent tectonic activity could expose them at the surface[60,61]. Within the crust, our experiment is most applicable to regimes where cryo-concentration is effective and produces localised brines[62,63]. Tidally-induced fracturing, leading to rapid ascent of ocean fluids, is the preferred explanation of how the plumes of Enceladus form[64,65]. Intrusive, melt-filled fracturing therefore may be expected within the crust of Enceladus where plume vents become sealed or where a fracture does not completely propagate to the surface[66]. We expect that our observation of natron is most analogous to these systems where localised cryo-concentrated brines may exist. Once formed crustal overturning could bring this natron-rich material to the surface.

Preservation of biosignatures in natron

To our knowledge, there have been no studies examining the preservation and viability of microbes within cryogenic salts. Inclusion formation is widely reported in evaporitic salt deposits[18,19]. As these salts precipitate from solution, defects and imperfections, or excessive under-cooling, can lead to irregular growth of the solid interface[18]. The timing of enclosure is largely kinetic, so these inclusions preserve a range of chemistries that track the evolution of the system.

Fluid inclusions are a rich source of geochemical information, as they trap fluid at the time of formation. Fluid inclusions in halite, a common evaporite mineral on Earth, have been used to reconstruct past ocean water physiochemistry[20]. More importantly, many salt-hosted fluid inclusions have been found with organic biological signatures[67,68] and even trapped microorganisms[23–25,69–71]. If microbial life is present in the ocean of Enceladus, then the precipitation of natron has the potential to trap life or signatures of life in fluid inclusions. These inclusions would then solidify at Enceladus surface temperatures.

Once present at or near the surface of Enceladus, natron may enhance the preservation of geochemical and biological signatures, if present. The higher density of natron (1.46 g cm^{-3})[72] compared to ice at 73 K (0.93 g cm^{-3})[73] would provide greater attenuation of energetic particles that would degrade organic signatures at the surface, partially shielding the inclusion contents[74]. Salt minerals have lower vapour pressure than ice[75], and therefore can be considered more structurally stable under vacuum at low temperatures. The limiting process is instead dehydration; Bu et al.[76] have shown that at Ceres-relevant equatorial temperatures (>200 K) natron will dehydrate within 6 days, but under Ceres polar conditions (<120 K), it will be stable for ~300 years. At Enceladus's mean surface temperature of ~70 K, natron should therefore be stable over long timescales, whereas in the anomalously warm terrain near the plume vents (140–225 K) dehydration may proceed within years. As the deposition and surface temperature will affect the long-term stability of natron, if signatures of natron are identified near the plume source, it is likely recently deposited and therefore worthy of investigation.

Missions

The plumes of Enceladus have already been proposed as high-priority for life detection experiments[77–79]. Our results suggest that remote observation should identify locations where crystalline natron is present as it could indicate slow-cooled, recently deposited material. Remote identification of crystalline natron therefore provides a direct means of selecting landing and sampling sites. We suggest that future missions prioritise searching for signs of crystalline natron in near-vent plume deposits as larger plume particles will be more likely to contain natron-hosted inclusions that we observe.

Realising natron's value as a sampling target requires first the ability to detect it. Natron has diagnostic hydration and carbonate absorption features in the near-infrared[80], yet no carbonate or chloride salt has been confirmed on Enceladus's surface as Cassini VIMS lacked the signal-to-noise to constrain salt abundance[45]. We therefore recommend that future missions carry near-infrared spectrometers with sufficient sensitivity to map hydrated carbonate, prioritising near-vent terrain where deposits are freshest and least dehydrated.

Conclusion

We have identified natron as an important solid component of the Na-Cl-CO_3-H_2O system at low temperatures. The freezing of an Enceladus-type Na-Cl-CO_3 solution generates natron crystals that contain primary fluid inclusions. As inclusions have the potential to trap biological signatures present in subsurface fluids, we suggest that natron is a mineral of astrobiological importance in the solar system. Future missions to Enceladus and other carbonate bearing icy bodies should have the capability to characterise surface mineralogy. This capability should then be used to identify sites that contain natron as a priority due to its potential to trap liquid and potentially signatures of life if they are present. Future work should explore the effects of cooling rate and composition on the formation of inclusions in cryogenic natron along with the potential for other low-solubility cryogenic salts to form inclusions.

Methods

Dual imaging and diffraction

Dual imaging (μCT) and diffraction (XRD) measurements were collected on the K11-DIAD beamline at Diamond Light Source, UK. DIAD's unique capabilities come from its two independently configurable X-ray beams allowing for image-guided, spatially-registered diffraction and imaging measurements (Supplementary Fig. 1).

An Na-Cl-CO_3-H_2O solution was synthesised gravimetrically using 1.0 mol kg^{-1} NaCl and 0.5 mol kg^{-1} Na_2CO_3 in 40 g of deionised (18 MΩ) water. The concentration of the solution we explored was approximately 5 times higher than estimates from plume-derived particles[7] but the ion ratios are comparable to plume measurements. An elevated concentration was chosen to ensure crystals of natron could be observed with imaging with sufficient resolution. The solution was then loaded into a 1 mm Kapton capillary. The temperature, controlled with a cryostream (Oxford Cryosystems 800 series), was first set to (20245) 0 K and then warmed to 270 K to allow for partial melting of the ice in the field of view; this annealing method allowed us to avoid issues with kinetics from nucleation inhibition. The solution was then step-cooled to 205 K at 6 K min^{-1} with tomography and diffraction measurements taken every 0.5 K up to 250 K and then every 5 K down to 205 K. 205 K was the lowest achievable temperature with this setup and ensured the solid system was quenched; a larger step size was used below 250 K as no obvious microstructural change was observed.

Full tomographs containing 1005 projections were collected across 180° with a monochromatic (20 keV) X-ray beam, and a 0.1 s exposure using a PCO-Edge 5.5 detector. The imaged field of view was 1.2 × 1.4 mm with the total reconstructed sample volume being 1.65 mm^3. The detector was binned twice to increase collection speed (102 s per scan), giving a pixel resolution of 2.16 μm per voxel. Ten dark and flat-field projections were

collected at the start and end to normalise the images before reconstruction. Imaging datasets were reconstructed using Filtered Back Projection (ASTRA-Toolbox) through SAVU [81].

Diffraction measurements were taken using a beam energy of 20 keV and a micro-focused spot size of 25 × 25 µm. 2D diffraction patterns were collected for 15 s using a Pilatus 2M area detector. DIAD's diffraction beam is collimated using a flat and bending mirror pair and the wavelength is selected using a double crystal monochromator. The beam is then directed and focused onto the sample position using a Kirkpatrick-Baez mirror system. This moving beam setup allows the diffraction measurements to be spatially registered with the imaging field of view and uniquely allows for image-guided diffraction. DIAD's geometry requires that only an arc of the Debye-Scherrer rings can be collected.

Data processing

Diffraction data were automatically azimuthally integrated and corrected for DIAD's moving beam geometry[82]. Pawley refinements for phase identification were carried out in TOPAS 7 Academic. Patterns were refined using parameters from published patterns[72,73,83].

Imaging datasets were pre-processed to improve the results of segmentation. Datasets were binned 2 times and both a bilateral filter and non-local means denoising filter were applied. Datasets were then segmented using a combination of a U-Net_plus_plus with pre-trained resnet50 encoders, using Volume Segmantics[84] and manual segmentation in Avizo. A single model was trained on 5 labelled volumes of 256 px^3 across the series of tomographies taken. BCEDiceLoss criteria were used with a DiceCoefficient as the metric. Size, shape, orientation and volume fractions were calculated using Avizo morphometric implementations. Surfaces were generated and simplified in Avizo and then rendered in Blender 3.0.

Data availability

All reconstructed tomography data; the UNET model and training data; and all reduced diffraction data from this study can be found in a zenodo repository [85].

Received: 11 February 2026; Accepted: 8 September 2026;
Published online: 06 October 2026

References

1. Porco, C. C. et al. Cassini observes the active south pole of Enceladus. *Science* **311**, 1393–1401 (2006).
2. Hansen, C. J. et al. Enceladus' water vapor plume. *Science* **311**, 1422–1425 (2006).
3. Crow-Willard, E. N. & Pappalardo, R. T. Structural mapping of Enceladus and implications for formation of tectonized regions. *J. Geophys. Res.: Planets* **120**, 928–950 (2015).
4. Dougherty, M. K. et al. Identification of a dynamic atmosphere at Enceladus with the Cassini magnetometer. *Science* **311**, 1406–1409 https://doi.org/10.1126/science.1120985 (2006).
5. Waite, J. H. et al. Cassini ion and neutral mass spectrometer: Enceladus plume composition and structure. *Science* **311**, 1419–1422 https://doi.org/10.1126/science.1121290 (2006).
6. Waite, J. H. et al. Cassini finds molecular hydrogen in the Enceladus plume: evidence for hydrothermal processes. *Science* **356**, 155–159 (2017).
7. Postberg, F., Schmidt, J., Hillier, J., Kempf, S. & Srama, R. A salt-water reservoir as the source of a compositionally stratified plume on Enceladus. *Nature* **474**, 620–622 (2011).
8. Khawaja, N. et al. Low-mass nitrogen-, oxygen-bearing, and aromatic compounds in Enceladean ice grains. *Mon. Not. R. Astron. Soc.* **489**, 5231–5243 (2019).
9. Postberg, F. et al. Detection of phosphates originating from Enceladus's ocean. *Nature* **618**, 489–493 (2023).
10. Thomas, P. C. et al. Enceladus's measured physical libration requires a global subsurface ocean. *Icarus* **264**, 37–47 (2016).
11. Nimmo, F., Spencer, J. R., Pappalardo, R. T. & Mullen, M. E. Shear heating as the origin of the plumes and heat flux on Enceladus. *Nature* **447**, 289–291 (2007).
12. Gall, A. L. et al. Thermally anomalous features in the subsurface of Enceladus's south polar terrain. *Nat. Astron.* **1**, 0063 (2017).
13. O'neill, C. & Nimmo, F. The role of episodic overturn in generating the surface geology and heat flow on Enceladus. *Nat. Geosci.* **3**, 88–91 (2010).
14. Neveu, M. et al. Returning samples from Enceladus for life detection. *Front. Astron. Space Sci.* **7**, 26 (2020).
15. Vincent, L. N., Fayolle, E. C., Hodyss, R., Johnson, P. V. & Noell, A. C. Bacterial spore morphology remains highly recognizable after exposure to simulated Enceladus and Europa surface conditions. *Commun. Earth Environ.* **5**, 688 (2024).
16. Choukroun, M. et al. Sampling plume deposits on Enceladus' surface to explore ocean materials and search for traces of life or biosignatures. *Planet. Sci. J.* **2**, 100 (2021).
17. Abramov, O. & Spencer, J. R. Endogenic heat from Enceladus' south polar fractures: new observations, and models of conductive surface heating. *Icarus* **199**, 189–196 (2009).
18. Roedder, E. *Fluid Inclusions* Vol. 12 (De Gruyter, 1984).
19. Goldstein, R. H. Fluid inclusions in sedimentary and diagenetic systems. *Lithos* **55**, 159–193 (2001).
20. Blättler, C. L. Examining uncertainties in major-ion reconstructions of seawater from halite fluid inclusions. *Earth Sci. Rev.* **267**, 105142 (2025).
21. Lowenstein, T. K., Timofeeff, M. N., Brennan, S. T., Hardie, L. A. & Demicco, R. V. Oscillations in Phanerozoic seawater chemistry: evidence from fluid inclusions. *Science* **294**, 1086–1088 (2001).
22. McDermott, F. Palaeo-climate reconstruction from stable isotope variations in speleothems: a review. *Quat. Sci. Rev.* **23**, 901–918 (2004).
23. McGenity, T. J., Gemmell, R. T., Grant, W. D. & Stan-Lotter, H. Origins of halophilic microorganisms in ancient salt deposits. *Environ. Microbiol.* **2**, 243–250 (2000).
24. Schreder-Gomes, S. I., Benison, K. C. & Bernau, J. A. 830-million-year-old microorganisms in primary fluid inclusions in halite. *Geology* **50**, 918–922 (2022).
25. Fang, Z., Knoll, P., McMahon, S., Qin, L. & Cockell, C. S. Preservation of microorganisms (Chroococcidiopsis sp. 029) in salt minerals under low atmospheric pressure: application to life detection on Mars. *Planet. Sci. J.* **5**, 263 (2024).
26. Lowenstein, T. K., Schubert, B. A. & Timofeeff, M. N. Microbial communities in fluid inclusions and long-term survival in halite. *GSA Today* **21**, 4–9 (2011).
27. Gibson, M. E. & Benison, K. C. It's a trap!: Modern and ancient halite as lagerstätten. *J. Sediment. Res.* **93**, 642–655 (2023).
28. Benison, K. C. How to search for life in martian chemical sediments and their fluid and solid inclusions using petrographic and spectroscopic methods. *Front. Environ. Sci.* **7**, 108 (2019).
29. Light, B., Brandt, R. E. & Warren, S. G. Hydrohalite in cold sea ice: laboratory observations of single crystals, surface accumulations, and migration rates under a temperature gradient, with application to "snowball earth". *J. Geophys. Res. Oceans* **114** https://doi.org/10.1029/2008JC005211 (2009).
30. Postberg, F. et al. Sodium salts in e-ring ice grains from an ocean below the surface of Enceladus. *Nature* **459**, 1098–1101 (2009).
31. McCoy, T. J. et al. An evaporite sequence from ancient brine recorded in Bennu samples. *Nature* **637**, 1072–1077 (2025).
32. Matsumoto, T. et al. Sodium carbonates on Ryugu as evidence of highly saline water in the outer solar system. *Nat. Astron.* **8**, 1536–1543 (2024).
33. Carrozzo, F. G. et al. Nature, formation, and distribution of carbonates on Ceres. *Sci. Adv.* **4** http://advances.sciencemag.org/ (2018).

34. Cartwright, R. J. et al. JWST reveals CO ice, concentrated CO_2 deposits, and evidence for carbonates potentially sourced from Ariel's interior. *Astrophys. J. Lett.* **970**, L29 (2024).
35. Marion, G. M. Carbonate mineral solubility at low temperatures in the Na-K-Mg-Ca-H-Cl-SO_4-OH-HCO_3-CO_3-CO_2-H_2O system. *Geochim. Cosmochim. Acta* **65**, 1883–1896 (2001).
36. Lécuyer, C. & O'neil, J. R. Stable isotope compositions of fluid inclusions in biogenic carbonates. *Geochim. Cosmochim. Acta* **58**, 353–363 (1994).
37. Glen, J. W. & Perutz, M. F. The growth and deformation of ice crystals. *J. Glaciol.* **2**, 397–403 (1954).
38. Cole, D. M. The microstructure of ice and its influence on mechanical properties. *Eng. Fract. Mech.* www.elsevier.com/locate/engfracmech (2001).
39. Pawley, G. S. Unit-cell refinement from powder diffraction scans. *J. Appl. Cryst.* **14**, 357–361 (1981).
40. Linke, W. F. *Solubilities: Inorganic and Metal-Organic Compounds* 4th edn Vol. II (American Chemical Society, 1958).
41. Hsu, H. W. et al. Ongoing hydrothermal activities within Enceladus. *Nature* **519**, 207–210 (2015).
42. Hurford, T. A., Helfenstein, P., Hoppa, G. V., Greenberg, R. & Bills, B. G. Eruptions arising from tidally controlled periodic openings of rifts on Enceladus. *Nature* **447**, 292–294 (2007).
43. Kite, E. S. & Rubin, A. M. Sustained eruptions on Enceladus explained by turbulent dissipation in tiger stripes. *Proc. Natl. Acad. Sci. USA* **113**, 3972–3975 (2016).
44. Hedman, M. M. et al. An observed correlation between plume activity and tidal stresses on Enceladus. *Nature* **500**, 182–184 (2013).
45. Postberg, F. et al. Plume and surface composition of Enceladus. In *Enceladus and the Icy Moons of Saturn* (The University of Arizona Press, 2018).
46. Spencer, J. R. et al. Cassini encounters Enceladus: background and the discovery of a south polar hot spot. *Science* **311**, 1401–1405 (2006).
47. Goguen, J. D. et al. The temperature and width of an active fissure on Enceladus measured with Cassini VIMS during the 14 April 2012 South Pole flyover. *Icarus* **226**, 1128–1137 (2013).
48. Ando, K., Arakawa, M. & Terasaki, A. Freezing of micrometer-sized liquid droplets of pure water evaporatively cooled in a vacuum. *Phys. Chem. Chem. Phys.* **20**, 28435–28444 (2018).
49. Klenner, F. et al. Supercooling, glass formation, and mineral assemblages upon freezing of salty ice grains from Enceladus's ocean. *Planet. Sci. J.* **6**, 65 (2025).
50. Imrichová, K. et al. Vitrification and increase of basicity in between ice Ih crystals in rapidly frozen dilute NaCl aqueous solutions. *J. Chem. Phys.* **151**, 014503 (2019).
51. Fox-Powell, M. G. & Cousins, C. R. Partitioning of crystalline and amorphous phases during freezing of simulated Enceladus ocean fluids. *J. Geophys. Res. Planets* **126**, e2020JE006628 (2021).
52. Hamp, R. E. et al. Metastable dihydrate of sodium chloride at ambient pressure. *J. Phys. Chem. Lett.* **15**, 12301–12308 (2024).
53. Meyer, C. R. et al. A potential mushy source for the geysers of Enceladus and other icy satellites. *Geophys. Res. Lett.* **52**, e2024GL111929 (2025).
54. Goldstein, D. B. et al. Enceladus plume dynamics: from surface to space. in *Enceladus and the icy Moons of Saturn* 175–194 (University of Arizona Press, 2018).
55. Dong, Y., Hill, T. W. & Ye, S. Y. Characteristics of ice grains in the Enceladus plume from Cassini observations. *J. Geophys. Res.: Space Phys.* **120**, 915–937 (2015).
56. Jaumann, R. et al. Distribution of icy particles across Enceladus' surface as derived from Cassini-VIMS measurements. *Icarus* **193**, 407–419 (2008).
57. Degruyter, W. & Manga, M. Cryoclastic origin of particles on the surface of Enceladus. *Geophys. Res. Lett.* **38** https://doi.org/10.1029/2011GL048235 (2011).
58. Rysgaard, S. et al. Ikaite crystal distribution in winter sea ice and implications for CO_2 system dynamics. *Cryosphere* **7**, 707–718 (2013).
59. čadek, O. et al. Long-term stability of Enceladus' uneven ice shell. *Icarus* **319**, 476–484 (2019).
60. Rossi, C., Cianfarra, P., Salvini, F., Bourgeois, O. & Tobie, G. Tectonics of Enceladus' South Pole: block rotation of the tiger stripes. *J. Geophys. Res. Planets* **125**, e2020JE006471 (2020).
61. Kargel, J. S. & Pozio, S. The volcanic and tectonic history of Enceladus. *ICARUS* **119**, 385–404 (1996).
62. Zolotov, M. Y. & Shock, E. L. Composition and stability of salts on the surface of Europa and their oceanic origin. *J. Geophys. Res. E Planets* **106**, 32815–32827 (2001).
63. Vance, S. D., Journaux, B., Hesse, M. & Steinbrügge, G. The salty secrets of icy ocean worlds. *J. Geophys. Res. Planets* **126**, e2020JE006736 (2021).
64. Spencer, J. et al. *Plume Origins and Plumbing (Ocean to Surface)* (The University of Arizona Press, 2018).
65. Ingersoll, A. P. & Nakajima, M. Controlled boiling on Enceladus. 2. Model of the liquid-filled cracks. *Icarus* **272**, 319–326 (2016).
66. Souček, O., Běhounková, M., Lanzendörfer, M., Tobie, G. & Choblet, G. Variations in plume activity reveal the dynamics of water-filled faults on Enceladus. *Nat. Commun.* **15**, 7405 (2024).
67. Shanina, S. N., Galamay, A. R., Ignatovich, O. O., Burdelnaya, N. S. & Valyaeva, O. V. Organic matter of the salt sequence in the southern part of the Yakshinskoe potassium-magnesium salt deposit. *Geochem. Int.* **56**, 719–734 (2018).
68. Pellegrino, L. et al. The impact of early diagenesis on biosignature preservation in sulfate evaporites: Insights from Messinian (Late Miocene) gypsum. *Geobiology* **22**, e70007 (2024).
69. Benison, K. C., Hallsworth, J. E., Zalar, P., Glavina, M. & Gunde-Cimerman, N. Extremophilic and common fungi in acid brines and their halite. *Extremophiles* **29**, 15 (2025).
70. Stan-Lotter, H. et al. Halococcus dombrowskii sp. nov., an archaeal isolate from a Permian alpine salt deposit. *Int. J. Syst. Evolut. Microbiol.* **52**, 1807–1814 (2002).
71. Gill, K. K., Jagniecki, E. A., Benison, K. C. & Gibson, M. E. A Mars-analog sulfate mineral, mirabilite, preserves biosignatures. *Geology* **51**, 818–822 (2023).
72. Taga, T. Crystal structure of $Na_2CO_3.10H_2O$. *Acta Crystallogr. Sect. B* **25**, 2656–2658 https://doi.org/10.1107/S0567740869006236 (1969).
73. Fortes, A. D. et al. No evidence for large-scale proton ordering in Antarctic ice from powder neutron diffraction. *J. Chem. Phys.* **120**, 11376–11379 (2004).
74. Kotova, A., Roussos, E., Kollmann, P., Krupp, N. & Dandouras, I. Galactic cosmic rays access to the magnetosphere of Saturn. *J. Geophys. Res.: Space Phys.* **124**, 166–177 (2019).
75. Chase, M. W. et al. JANAF thermochemical tables—3rd edition .1. Al-Co. *J. Phys. Chem. Ref. Data* **14**, 1–926 (1985).
76. Bu, C. et al. Stability of hydrated carbonates on Ceres. *Icarus* **320**, 136–149 (2019).
77. Porco, C. C., Dones, L. & Mitchell, C. Could it be snowing microbes on Enceladus? assessing conditions in its plume and implications for future missions. *Astrobiology* **17**, 876–901 (2017).
78. Perera, L. J. & Cockell, C. S. Dispersion of bacteria by low-pressure boiling: life detection in Enceladus' plume material. *Astrobiology* **23**, 269–279 (2023).
79. Klenner, F. et al. How to identify cell material in a single ice grain emitted from Enceladus or Europa. *Sci. Adv.* **10**, 849 (2024).
80. Angelis, S. D. et al. NIR reflectance spectroscopy of hydrated and anhydrous sodium carbonates at different temperatures. *Icarus* **317**, 388–411 (2019).

81. Wadeson, N. & Basham, M. Savu: A python-based, MPI framework for simultaneous processing of multiple, n-dimensional, large tomography datasets. https://doi.org/10.48550/arXiv.1610.08015 (2016).
82. Leonardi, A. et al. The moving-beam diffraction geometry: the DIAD application of a diffraction scanning probe. *J. Appl. Crystallogr.* **59**, 12–27 (2026).
83. Klewe, B. & Pedersen, B. The crystal structure of sodium chloride dihydrate. *Acta Crystallogr. Sect. B* **30**, 2363–2371 (1974).
84. King, O. N. F., Bellos, D. & Basham, M. Volume segmantics: a Python package for semantic segmentation of volumetric data using pre-trained pytorch deep learning models. *J. Open Source Softw.* **7**, 4691 (2022).
85. Perera, L. J. & Houx, J. L. Spatially resolved synchrotron X-ray tomography and diffraction of cryogenic natron and fluid inclusions in a frozen Na-Cl-CO3 solution (2026).

Acknowledgements

The authors would like to acknowledge Diamond Light Source for access to the DIAD beamline under proposal MG31838 and MG38841.

Author contributions

Conceptualisation: L.J.P. Methodology: L.J.P., J.L.H., S.J.D., S.P.T. Investigation: L.J.P., J.L.H., A.L., S.J.D., S.P.T. Visualisation: L.J.P. Writing: L.J.P., J.L.H., A.L., S.J.D,. S.P.T.

Funding

JLH was partially funded by the Rutherford Appleton Laboratory and Faraday Institution through his emerging leader fellowship, FIELF001, and by Research England's 'Expanding Excellence in England' grant through the "Multi-scale Multi-disciplinary Modelling for Impact" program ($M^3$4Impact).

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43247-026-04080-z.

Correspondence and requests for materials should be addressed to Liam J. Perera.

Peer review information *Communications Earth & Environment* the anonymous reviewers for their contribution to the peer review of this work. Primary Handling Editors: NikFita Kaushal and Joseph Aslin. A peer review file is available.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

© The Author(s) 2026

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.